

HALF-LIFE SIMULATION

Aim: To simulate radioactive decay and to determine the half-life of this simulated decay.

Hypothesis:

Apparatus: 1 cup, 25 M&Ms, clean paper

Method:

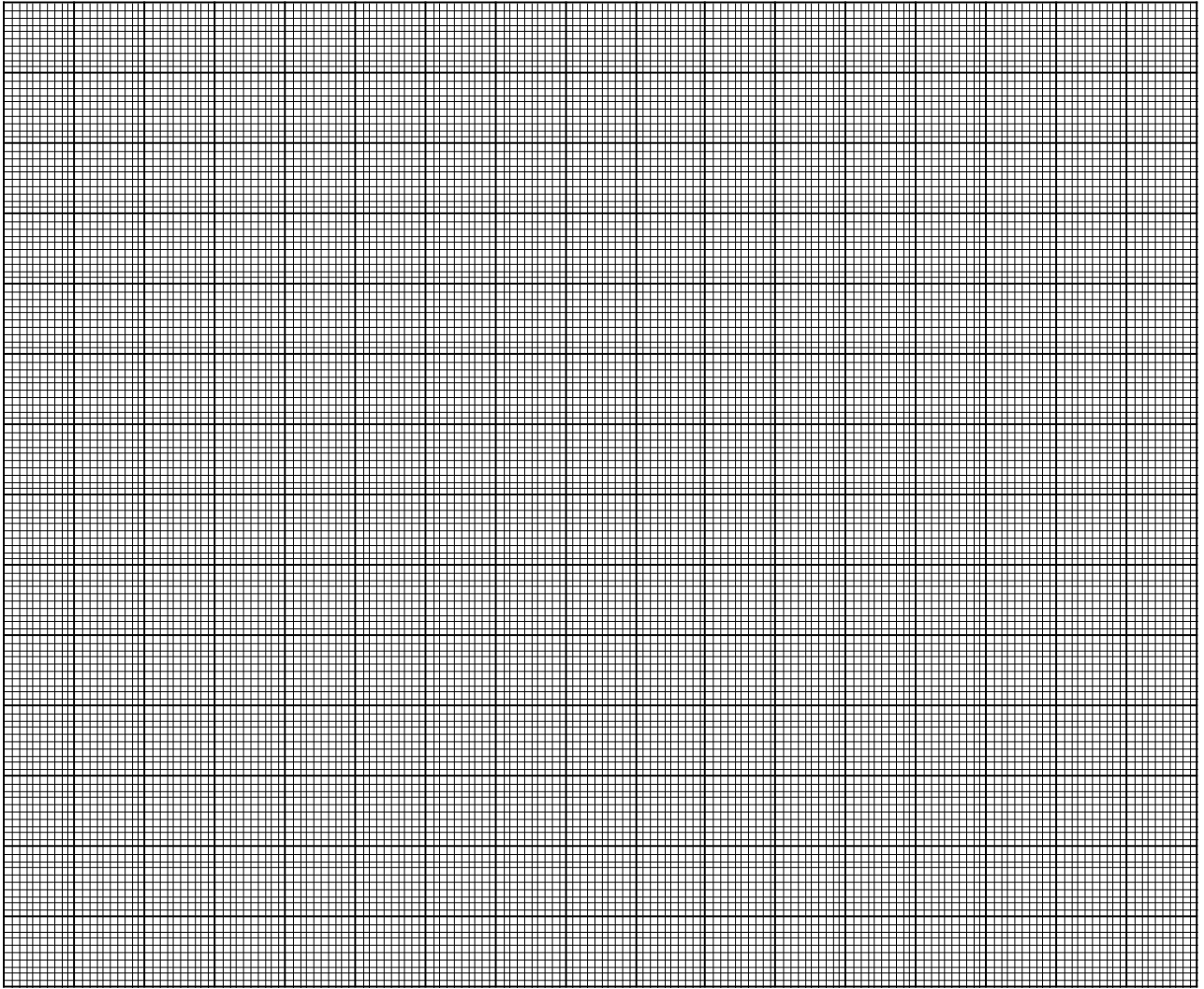
1. Working in pairs, collect your M&Ms and place them in your plastic cup. Write the number of M&Ms into the table for number of throws = 0.
2. Tip the M&Ms out onto a piece of paper.
3. Count and remove the "decayed" nuclei (the M&Ms that land with the M side down) and record this value in your table as the number decayed. Remove these M&Ms.
4. Place the remaining M&Ms (those that landed M side up) back into your cup.
5. Repeat steps 2 - 4, until all of the M&Ms have decayed.
6. Graph your results as: number of nuclei remaining against number of throws.
7. Determine the half-life of your simulation.

Note: If time permits, you may do a second simulation.

Your class teacher may also collate data from every group to get the class results.

Results:

Number of Throws	Number of Disintegrations	Number of Nuclei Remaining
0	0	
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
13		
14		
15		



Discussion:

Conclusion: